

Rezolvarea ec. diferențiale cu Transformata Laplace

10.11.2025

1. Se aplică T.L. pe fiecare element în parte din ec. diferențială. Se obține $Y(s)$
2. Se aplică T.L.i.

$$\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = 3u(t)$$

$$y(0) = a ; y'(0) = b , u(t) = 1_+(t)$$

$$\mathcal{L}\{u(t)\} = \frac{1}{s} ; \mathcal{L}\{y(t)\} = Y(s)$$

$$\mathcal{L}\left\{\frac{dy(t)}{dt}\right\} = sY(s) - y(0) = sY(s) - a$$

$$\mathcal{L}\left\{\frac{d^2 y(t)}{dt^2}\right\} = s^2 Y(s) - sy(0) - y'(0) = s^2 Y(s) - sa - b$$

$$s^2 Y(s) - sa - b + 3sY(s) - 3a + 2Y(s) = \frac{3}{s}$$

$$Y(s)(s^2 + 3s + 2) = sa + b + 3a + \frac{3}{s}$$

$$Y(s) = \frac{s^2 a + b + 3a + 3}{s(s^2 + 3s + 2)} = \frac{s^2 a + b + 3a}{s(s+1)(s+2)} + \frac{3}{s(s+1)(s+2)}$$

$$Y_m(s) = \frac{sa + b + 3a}{(s+1)(s+2)} = \frac{C_1}{s+1} + \frac{C_2}{s+2}$$

$$C_1 = (s+1)Y_m(s) \Big|_{s=-1} = \frac{(-1+2)(-a+b+3a)}{(-1+2)(-1+2)} = 2a+b$$

$$C_2 = (s+2) Y_m(s) \big|_{s=-2} = -(a+b)$$

$$Y_m(s) = \frac{2a+b}{s+1} + \frac{-(a+b)}{s+2} \bigg| \mathcal{L}^{-1} \Rightarrow y_m(t) = (2a+b)e^{-t} - (a+b)e^{-2t}$$

$$Y_f(s) = \frac{3}{s(s+1)(s+2)} = \underbrace{\frac{C_3}{s}}_{Y_{st}(s)} + \underbrace{\frac{C_4}{s+1} + \frac{C_5}{s+2}}_{Y_{tr}(s)}$$

$$C_3 = s Y_f(s) \big|_{s=0} = \frac{3}{2}$$

$$C_4 = (s+1) Y_f(s) \big|_{s=-1} = -3$$

$$C_5 = (s+2) Y_f(s) \big|_{s=-2} = \frac{3}{2}$$

$$Y_f(s) = \frac{3}{2s} + \frac{3}{s+1} + \frac{3}{2} \frac{1}{s+2} \bigg| \mathcal{L}^{-1} \Rightarrow y_f(t) = \frac{3}{2} 1_+(t) - 3e^{-t} + \frac{3}{2} e^{-2t}$$

$$\frac{3}{2} e^{-2t}; t \geq 0$$

$$y(t) = (2a+b)e^{-t} - (a+b)e^{-2t} + \frac{3}{2} 1_+(t) - 3e^{-t} + \frac{3}{2} e^{-2t}; t \geq 0$$

$$\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + 5y(t) = 2u(t)$$

$$\mathcal{L}\{u(t)\} = \frac{1}{s^2}$$

$$\mathcal{L}\{y(t)\} = Y(s)$$

$$\mathcal{L}\{y'(t)\} = sY(s) + y(0) = sY(s)$$

$$\mathcal{L}\left\{\frac{d^2 y(t)}{dt^2}\right\} = s^2 Y(s) + sy(0) + y'(0) = s^2 Y(s)$$

$$s^2 Y(s) + 2sY(s) + 5Y(s) = \frac{2}{s^2}$$

$$Y(s) = \frac{2}{s^2(s^2+2s+5)} = \frac{C_1}{s^2} + \frac{C_2}{s} + \frac{C_3s+C_4}{s^2+2s+5}$$

$$C_1 = s^2 Y(s) \Big|_{s=0} = \frac{2}{s^2+2s+5} \Big|_{s=0} = \frac{2}{5}$$

$$C_2 = \frac{dC_1}{ds} \Big|_{s=0} = \frac{-2(2s+2)}{(s^2+2s+5)^2} \Big|_{s=0} = \frac{-4}{25}$$

$$C_3 = \frac{4}{25}$$

$$C_4 = \frac{-2}{25}$$

$$Y(s) = \frac{2}{5} \frac{1}{s^2} + \frac{-4}{25} \frac{1}{s} + \frac{C_3s+C_4}{s^2+2s+5}$$

$$Y(s) = \frac{2}{5s^2} - \frac{4}{25s} + \frac{C_3s+C_4}{s^2+2s+5}$$

$$Y(s) = \frac{50s - 20s^2}{125s^3} + \frac{C_3s+C_4}{s^2+2s+5}$$

$$Y(s) = \frac{(50s - 20s^2)(s^2+2s+5) + (C_3s+C_4)125s^3}{125s^3(s^2+2s+5)}$$

$$Y(s) = \frac{2}{5} \frac{1}{s^2} - \frac{4}{25} \frac{1}{s} + \frac{\frac{4}{25}s - \frac{2}{25}}{s^2 + 2s + 5} = \frac{2}{5} \frac{1}{s^2} - \frac{4}{25} \frac{1}{s} + \frac{4}{25} \frac{s - \frac{1}{2}}{(s+1)^2 + 2^2}$$

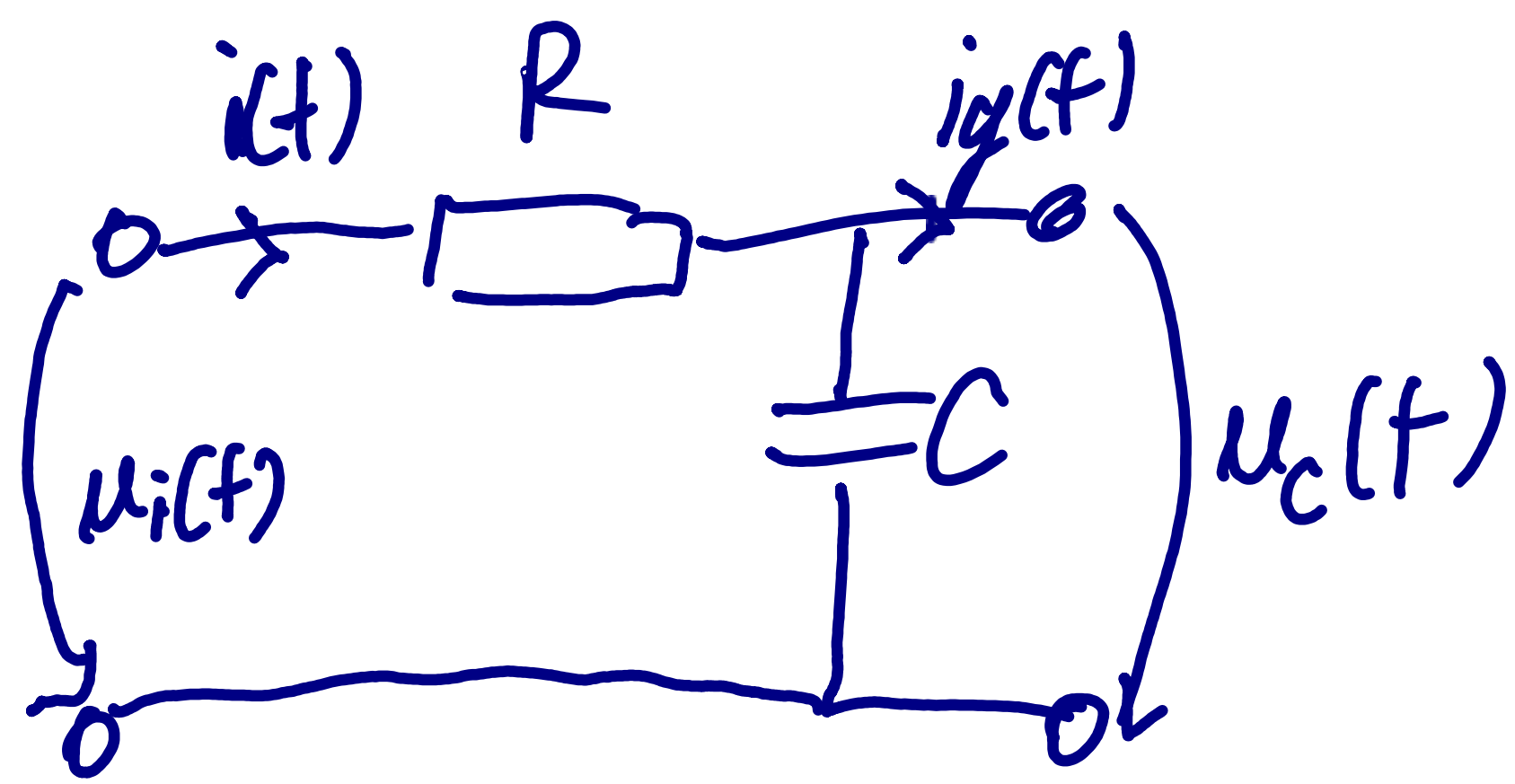
$$e^{-at} \sin \omega t = \frac{\omega}{(s+a)^2 + \omega^2}$$

$$e^{-at} \cos \omega t = \frac{s+a}{(s+a)^2 + \omega^2}$$

$$= \frac{2}{5} \frac{1}{s^2} - \frac{4}{25} \frac{1}{s} + \frac{4}{25} \frac{s+1}{(s+1)^2 + 2^2} + \frac{4}{25} \frac{-\frac{3}{2}}{(s+1)^2 + 2^2}$$

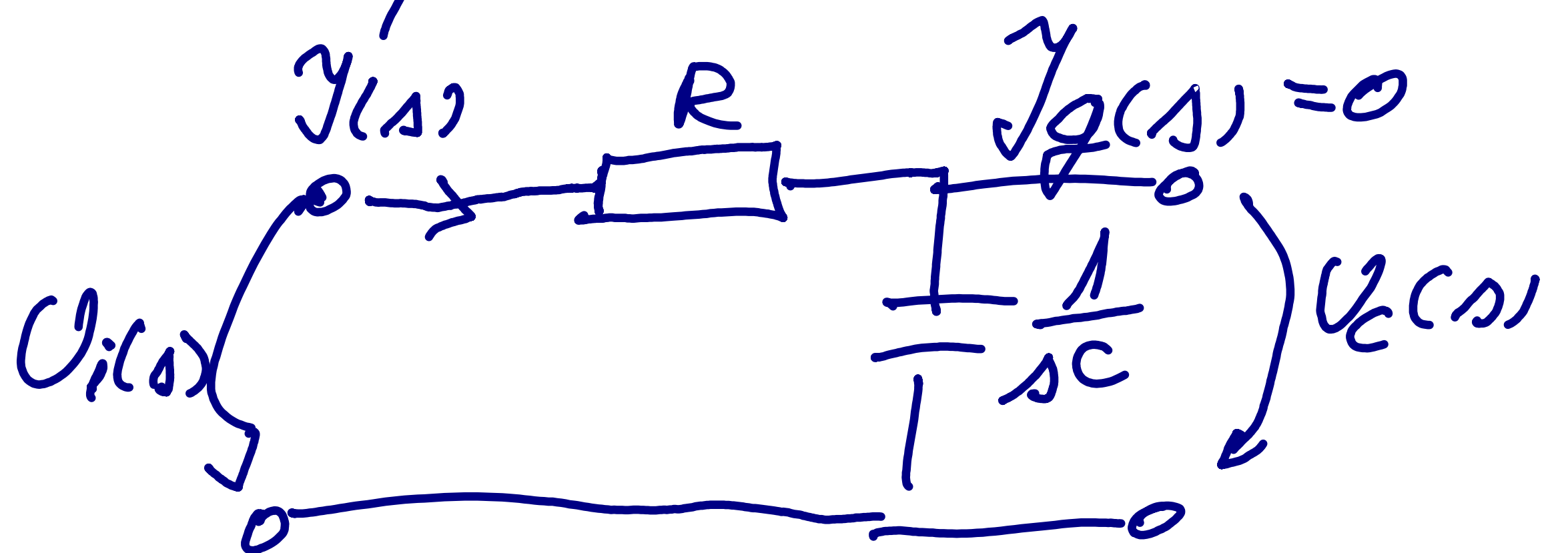
$$y(t) = \frac{2}{5}t - \frac{4}{25}1_+(t) + \frac{4}{25}e^{-t} \cos 2t - \frac{3}{25}e^{-t} \sin 2t$$

Obținerea funcțiilor de transfer:



$$u(t) = u_i(t)$$

$$y(t) = u_c(t)$$



$$U(s) = U_i(s) \quad Z_1 = R$$

$$Y(s) = U_c(s) \quad Z_2 = \frac{1}{sC}$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{U_c(s)}{U_i(s)} = \frac{Y(s) \frac{1}{sC}}{Y(s) Z_e}$$

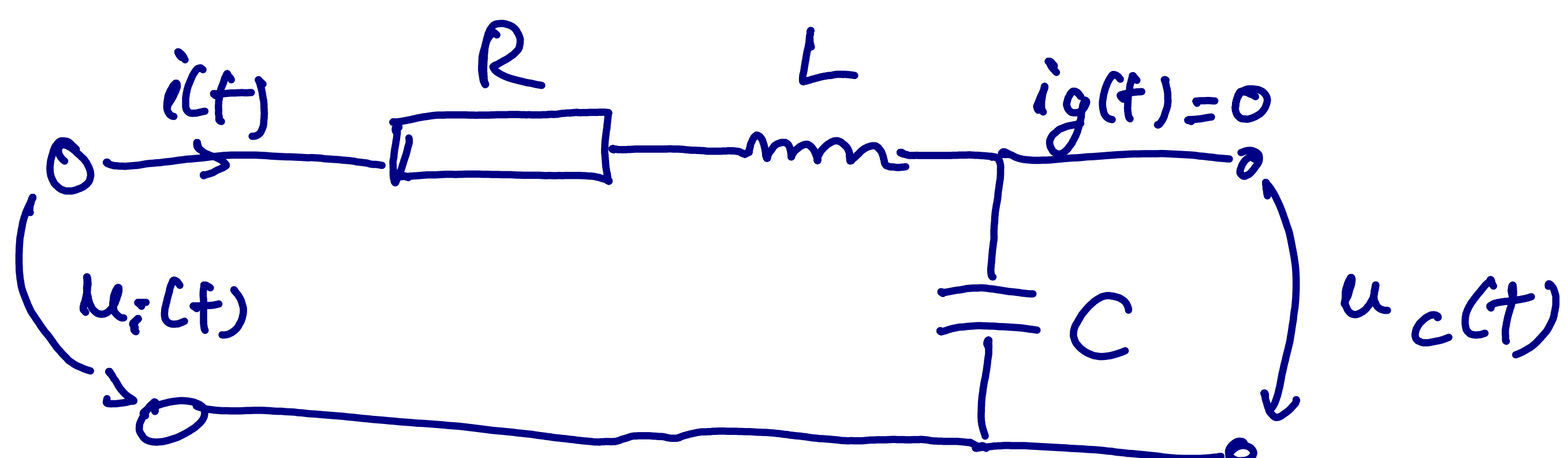
$$Z_e = Z_1 + Z_2$$

$$y(0) = 0$$

$$RC(sY(s) - y(0)) + Y(s) = U(s)$$

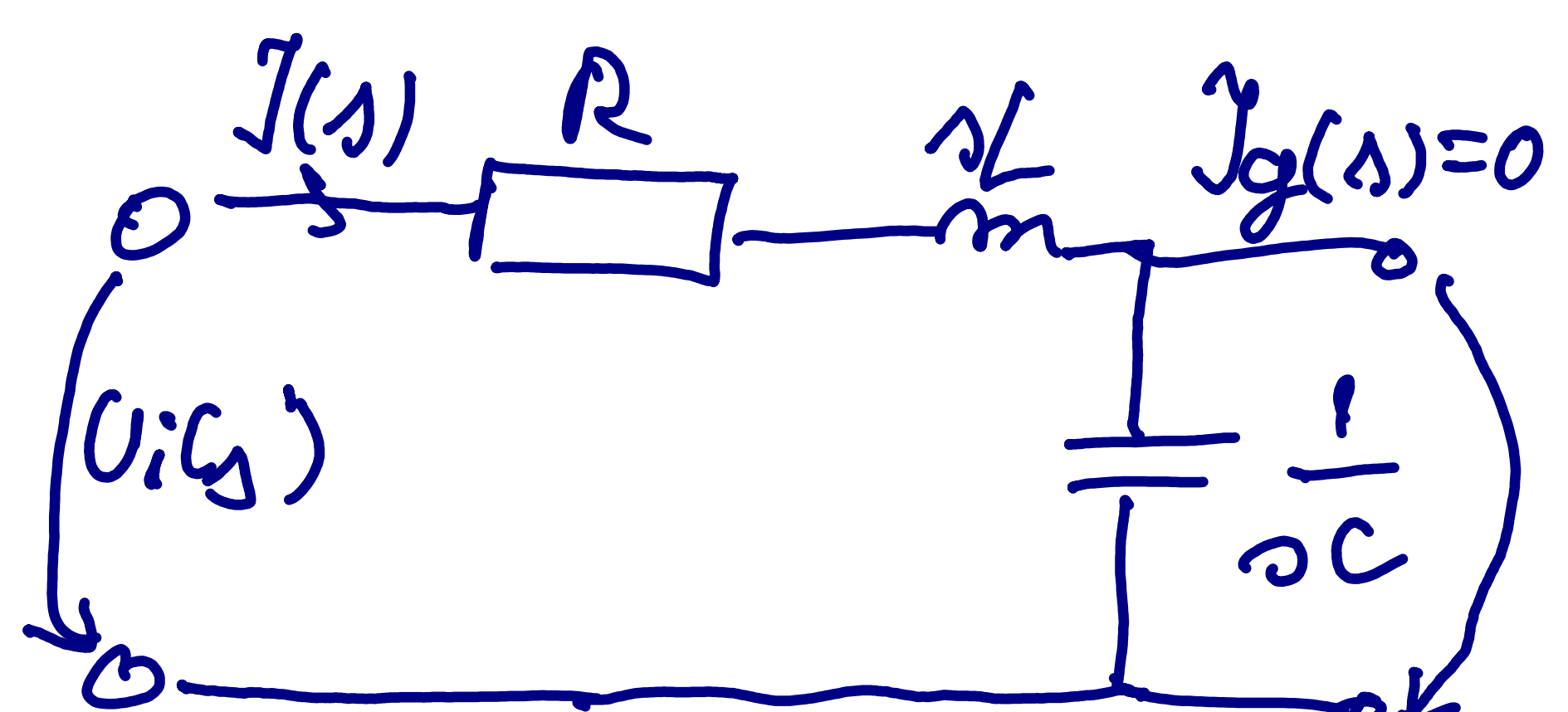
$$G(s) = \frac{Y(s)}{U(s)} = \frac{1}{RCs + 1}$$

$$G(s) = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{1}{RCs + 1}$$



$$u(t) = u_i(t)$$

$$y(t) = u_c(t)$$

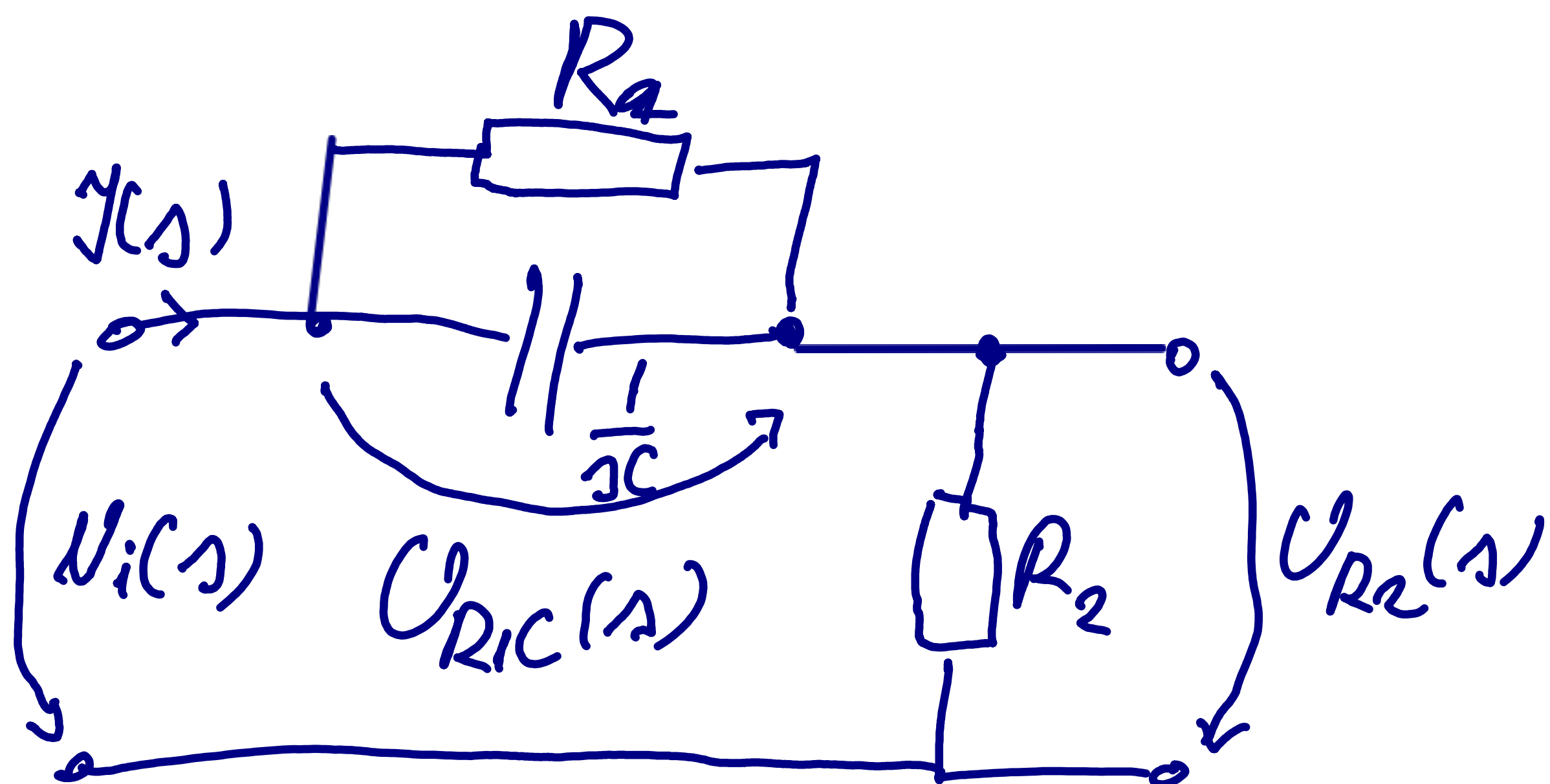
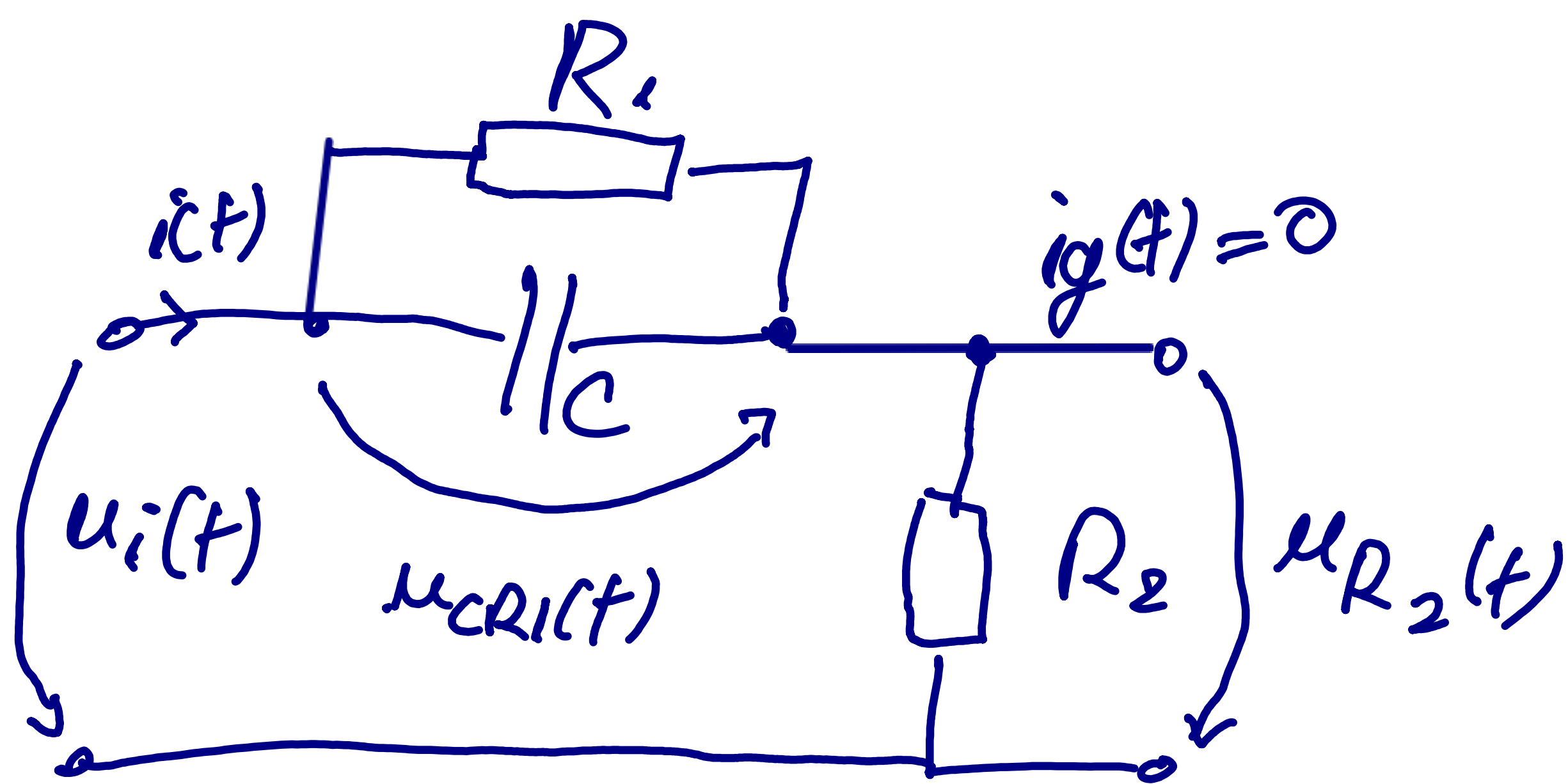


$$U(s) = U_i(s)$$

$$Y(s) = U_c(s)$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{I(s)z_3}{I(s)z_e} =$$

$$= \frac{\frac{1}{sC}}{R + sL + \frac{1}{sC}} = \frac{1}{RCs + LCs^2 + 1}$$



$$\frac{1}{z_e} = \frac{1}{z_1} + \frac{1}{z_2} \Rightarrow z_e = \frac{1}{RCs + 1} + z_3$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{I(s)z_3}{I(s)z_e} = \frac{R_2}{R_2 + \frac{R_1}{RCs + 1}}$$